



2026

METHODOLOGY OF Food Balance Sheet Statistics

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1. Introduction

Food Balance Sheet statistics are a fundamental tool for providing a clear and accurate picture of the balance between food production and consumption in the Emirate of Abu Dhabi. This is achieved by examining the quantities of food available in the markets and comparing them with the actual needs of the population. These statistics contribute to monitoring developments in food availability and its various sources, thereby reflecting the reality of food consumption at the individual and household levels.

Food Balance Sheet statistics are of particular importance in assessing the level of food security in the Emirate, as they help identify potential gaps between local food production and consumption levels, in addition to measuring the degree of reliance on external sources. They also enable a better understanding of food consumption patterns and the adequacy of intake in terms of essential nutrients.

Through these statistics, government entities, decision-makers, and planners are able to make decisions based on accurate and reliable data. This supports the development of agricultural and food policies, enhances the efficiency of resource allocation, and contributes to achieving sustainable food balance and ensuring the long-term availability of sufficient quantities of high-quality food.

2. Statistical Information

2.1. Importance and Objectives

Food Balance Sheet statistics serve as a key reference for understanding the status of food security and consumption patterns in the Emirate of Abu Dhabi, playing an important role in evaluating various food and agricultural policies and programs.

The main objectives of analysing Food Balance Sheet statistics include:

- Providing accurate and reliable statistical data and information to stakeholders in the food and food security sector.
- Monitoring indicators of food availability and consumption.
- Supplying decision-makers and government entities with the necessary data to assess the community's food needs and develop future plans.
- Supporting decision-makers in the development and direction of food and agricultural policies.
- Understanding the nutritional status of the population and enhancing the sustainability of food security

2.2. Key Outputs

Food Balance Sheet statistics provide an integrated framework comprising 15 statistical indicators that describe the reality of individual food consumption in the Emirate of Abu Dhabi. This is achieved by measuring the average per capita intake of food and key nutritional elements, including energy (calories), proteins, and fats, specifying their plant and animal sources and classifying them by household type. These indicators also reflect the quantities of food consumed on a daily or annual basis, offering a comprehensive understanding of food consumption patterns and their balance. These indicators are used to assess food adequacy and quality, analyse consumption patterns, and measure the level of food security. This, in turn, supports informed decision-making, the formulation of food and nutrition policies, and the enhancement of food security sustainability. The list of indicators is provided in the attached Table (1).

2.3. Concepts and definitions

Clear and precise concepts and definitions ensure consistency in interpretation and application across various Food Balance Sheet statistics indicators. These concepts form the foundation for data collection and analysis, supporting a unified approach to assessing the characteristics of these groups and analyzing their conditions. The concepts and definitions adopted by the Statistics Center – Abu Dhabi are aligned with internationally recognized terminology. They include the following:

- **Per capita Fat:**

It is an estimation of the available supply per capita of Fats which derived from the total food supplies available for human consumption from plant and animals' products (Source: Guidelines for the compilation of Food Balance Sheets - Food and Agriculture Organization of the United Nations) .

- **Per capita protein:**

It is an estimation of the available supply per capita of protein which derived from the total food supplies available for human consumption from plant and animals' products (Source: Guidelines for the compilation of Food Balance Sheets - Food and Agriculture Organization of the United Nations) .

- **Per capita food:**

It is an estimation of per capita food supplies available for human consumption in the reference period from plant and animals' products (Source: Guidelines for the compilation of Food Balance Sheets - Food and Agriculture Organization of the United Nations) .

- **Per capita calories:**

It is an estimation of the available supply per capita of calories which derived from the total food supplies available for human consumption from plant and animals' products (Source: Guidelines for the compilation of Food Balance Sheets - Food and Agriculture Organization of the United Nations) .

- **The food consumption patterns:**

The food consumption patterns reflect what either individuals or groups consume in terms of solid foods, beverages, including drinking water, and food supplements. (Source: Guidelines for the compilation of Food Balance Sheets - Food and Agriculture Organization of the United Nations).

2.4. Classifications and Standards applied

The purpose of using these sources is to ensure the adoption of accurate and standardized terms and criteria in the classification and documentation of Food Balance Sheet data, thereby ensuring comparability and alignment with best local and international practices.

- **FAO Guidelines for the Compilation of Food Balance Sheets¹:**

The Food Balance Sheet compilation guidelines provide a set of rules and standards for defining, classifying, and standardizing food expenditure data, ensuring accuracy, comparability, and consistency across different entities and time periods. These guidelines aim to support the preparation of reliable statistics on food consumption, enhance planning efficiency, and facilitate data-driven decision-making.

- **Reference Data for the Emirate of Abu Dhabi²:**

A strategic project at the Statistics Centre – Abu Dhabi, aiming to establish a unified, systematic, and technically efficient framework for the governance and management of reference data across the Emirate. It seeks to provide a standardized basis for data collection, classification, organization, and analysis. This initiative contributes to enhancing data integration, comparability, and the production of accurate and reliable statistical indicators, while promoting collaboration among all stakeholders within the Emirate's data ecosystem.

2.5. Available Breakdown

The Statistics Center - Abu Dhabi - publishes a wide range of indicators related to Food Balance Sheet statistics. These are published according to the following levels:

Average daily per capita fats:

- Household type
- Plant-based products
- Animal-based products

Average daily per capita protein:

- Household type
- Plant-based products
- Animal-based products

¹ <https://www.fao.org/3/ca6404en/ca6404en.pdf>

² <https://scad.gov.ae/web/guest/abu-dhabi-reference-classifications>

Average annual per capita foods:

- Household type
- Plant-based products
- Animal-based products

Average daily per capita calories from plant production:

- Household type

Average daily per capita calories from animal production.**Daily per capita energy, proteins and fats:**

- Household type

Food consumption patterns.**The main variables are also classified according to:****Household type:**

- Emirati Household
- Non-Emirati Household

Per capita:

- Energy
- Proteins
- Fats
- Foods

2.6. Statistical Population

The statistical population for the Food Balance Sheet statistics includes all residents of the Emirate of Abu Dhabi.

2.7. Statistical Unit

The statistical units adopted in this methodology include all individuals in the Emirate of Abu Dhabi.

2.8. Geographical Coverage

Food Balance Sheet statistics cover the three regions of the Emirate of Abu Dhabi: Abu Dhabi, Al Ain, and Al Dhafra.

2.9. Unit of Measure

Food Balance Sheet statistics are presented using specific units of measurement appropriate to the nature of each indicator, ensuring data clarity and ease of analysis. The main units include:

- **Kilogram (kg):** This unit is used to measure the average annual per capita food consumption.
- **Kilocalorie (kcal):** This unit is used to measure other food balance indicators, specifically the average daily per capita intake of fat, protein, calories, and energy.

2.10. Reference Period

The reference period for the data of all indicators is the survey year.

2.11. Timeliness

Food Balance Sheet statistics indicators are released within twelve months after the most recent reference period for which data are available, except for the Food Consumption Pattern indicator, which is released within seven months after the most recent reference period for which data are available.

2.12. Frequency of Dissemination

Food Balance Sheet statistics are published every five years.

2.13. Data Time Series

The length of the time series for available Food Balance Sheet statistics data varies by indicator, covering different periods that reflect the development of data collection and the expansion of indicators over the years. The time series starts from 2007 and is available at five-year intervals.

2.14. Abu Dhabi Special Considerations

Data collection is limited to the territorial boundaries of the Emirate of Abu Dhabi. However, the administrative data sources used in this process are also structured to operate at the emirate level, ensuring data integrity and reliability.

3. Statistical Processing

3.1. Data Sources

Food balance sheet statistics are compiled using data from the following statistical surveys: the Household Income and Expenditure Survey, in addition to population census data, which are used to calculate indicators related to population size.

3.1.1. Survey Data

The Household Income and Expenditure Survey is the primary source for estimating food balance sheet statistics, as it provides detailed data on the quantities and values of foods consumed by households during a specified reference period, whether purchased, self-produced, or received in kind. These data are used to calculate per capita averages of food commodities and their nutritional components, after applying the necessary statistical processing, such as converting quantities into edible equivalents and linking them to food composition tables. This enables the derivation of food balance sheet indicators and the assessment of dietary consumption patterns and levels of food availability.

3.1.2. Administrative Data

Some indicators are based on population data from the register-based census and are calculated by the Statistics Centre – Abu Dhabi.

3.2. Data Validation and Editing

3.2.1. Data Validation

The process of validating and auditing raw data is of critical importance, as it forms the foundation for subsequent statistical analysis. The Statistics Centre – Abu Dhabi applies various measures to ensure data accuracy and high-quality standards. This includes verifying that all required data are complete and meet the Centre's requirements and regularly auditing administrative data to ensure that errors are detected, addressed, and properly documented. Data consistency with statistical definitions and classifications is also ensured, extreme values are reviewed, and data plausibility is assessed. Food Balance sheet data are reviewed and audited in relation to previous years and related datasets, with logical validation conducted by examining both internal consistency (relationships, structure, and linkages within the dataset) and external consistency (data provided in previous years and other related data). In some cases, automated rules embedded in electronic data collection platforms are also used to detect errors or identify unacceptable responses or entries. Error reports are then communicated to the relevant stakeholders.

In certain cases, the Statistics Centre may also carry out—or request government entities to carry out—the following types of data checks:

- **Data validity and completeness:** Data validity refers to the correctness of responses or entries based on the permissible options (numbers, percentages, etc.).
- **Range checks:** Define the minimum and maximum expected values for a variable.
- **Duplicate data entry:** Verification that data for each unit in the register or database are entered only once, without duplication, particularly when discrepancies exist in some indicator fields for the same unit within the record.
- **Logical consistency:** Consistency refers to the existence of logical relationships and coherence between variables. This type of editing considers the linkages between data fields or variables.
- **Outliers:** This type of check follows other validation procedures and is used to detect outlier values based on the distribution of current data and historical data series, facilitating the identification of values that may be considered unusual or extreme, so they can be examined and verified.

3.2.2. Missing Data Adjustments

Handling missing data is a fundamental step in ensuring the quality and accuracy of food balance sheet statistical indicators. Data is thoroughly reviewed during collection and processing, and corrective action is taken if deficiencies or inconsistencies are found, either by contacting the source or by verifying relevant time series and classifications. This approach helps to minimize gaps and ensure the comprehensiveness of records, thereby enhancing the reliability of the results obtained from this methodology.

3.3. Statistical Calculation Method

The computational methods used in Food Balance Sheet statistics ensure accuracy and consistency in data analysis and reporting. These calculations include the following:

- **Average annual per capita food availability by household type:**

$[(\text{Quantity of food available from plant production} + \text{Quantity of food available from animal production}) \times 1000] / \text{Total population}$], broken down by household type.

1000: To convert the quantity from tons to kilogram

- **Average annual per capita food availability from plant products:**

$[(\text{Quantity of food available from plant production} \times 1000) / \text{Total population}]$.

1000: To convert the quantity from tons to kilogram

- **Average annual per capita food availability from animal products:**

$[(\text{Quantity of food available from animal production} \times 1000) / \text{Total population}]$

1000: To convert the quantity from tons to kilogram

- **Average daily per capita fat intake by household type:**

$[(\text{Total quantity of fat produced from plant production} + \text{Total quantity of fat produced from animal production}) / (\text{Total population} \times 365)]$, broken down by household type

- **Average daily per capita fat intake from plant products:**

$[\text{Total quantity of fat produced from plant production} / (\text{Total population} \times 365)]$

- **Average daily per capita fat intake from animal products:**

$[\text{Total quantity of fat produced from animal production} / (\text{Total population} \times 365)]$

- **Average daily per capita protein intake by household type:**

$[(\text{Total quantity of proteins produced from plant production} + \text{Total quantity of proteins produced from animal production}) / (\text{Total population} \times 365)]$, broken down by household type

- **Average daily per capita protein intake from plant products:**

$[\text{Total quantity of proteins produced from plant production} / (\text{Total population} \times 365)]$

- **Average daily per capita protein intake from animal products:**

[Total quantity of proteins produced from animal production / (Total population x 365)]

- **Average daily per capita calorie intake from plant products by household type:**

[Total quantity of calories produced from plant production / (Total population x 365)], broken down by household type

- **Average daily per capita calorie intake from plant products:**

[Total quantity of calories produced from plant production / (Total population x 365)]

- **Average daily per capita calorie intake from animal products:**

[Total quantity of calories produced from animal production / (Total population x 365)]

- **Daily per capita intake of energy, proteins, and fats:**

[(Total quantity of calories produced from plant + Total quantity of calories produced from animal production) / (Total population x 365)]

[(Total quantity of proteins produced from plant + Total quantity of proteins produced from animal production) / (Total population x 365)]

[(Total quantity of fat produced from plant + Total quantity of fat produced from animal production) / (Total population x 365)]

3.4. Seasonal Adjustment

The Statistics Centre – Abu Dhabi does not apply seasonal adjustments to food balance sheet statistics. This decision is based on the understanding that food balance sheet statistics are not significantly affected by seasonal adjustments.

4. Methodology Changes

Methodological changes related to administrative data may occur in line with changes in quality assurance and auditing processes, organizational structures, and applicable systems. The methodology may also be adjusted taking into account factors such as policy objectives and the relevance of the indicator. However, no recent methodological changes have been made with respect to food balance sheet statistics.

5. Revision Policy

Continuous data refinement is mandated to ensure the accuracy of data and statistical products provided by the Statistics Centre – Abu Dhabi. This involves revising official statistics published through printed documents or approved dissemination channels. The policy outlines general rules for reviewing and publishing statistics, informing users of potential review reasons, categorizing refinements, and documenting all review aspects. This framework allows producers to determine review mechanisms based on specific characteristics. The refinement policy covers all indicators, statistics, and time series of varying periodicities. Additionally, the Statistics Centre – Abu Dhabi publishes a general data refinement policy applicable to all statistics under its responsibility, ensuring adaptability to the distinct characteristics of various statistics and allowing for specialized refinement policies in alignment with the general policy.

6. Statistical Quality

The quality of statistical data focuses on the concept of (data relevance for use) by preparing statistical data characterized by a set of traits and properties that meet and align with the data requirements of users and decision-makers. Thus, statistical quality exists within an institutional environment that is characterized by professionalism, integrity, and neutrality, closely related to the set objectives for the data and the needs of users. This is achieved through the use of a clear, standardized framework that ensures the data's reliability, realism, comparability, efficiency, accessibility, and flexibility, and that it is available to all within an appropriate timeframe and in an easily accessible and usable manner. Statistical product quality reports are prepared by the Quality Team in SCAD.

7. Appendix Table

S.V. Code	Indicator Name	اسم المؤشر
2324	Average annual per capita foods by household type	متوسط نصيب الفرد السنوي من الغذاء حسب نوع الأسرة
2364	Average annual per capita foods from animal production	متوسط نصيب الفرد السنوي من الغذاء من المنتجات الحيوانية
2323	Average annual per capita foods from plant production	متوسط نصيب الفرد السنوي من الغذاء من المنتجات النباتية
2322	Average daily per capita protein by household type	متوسط نصيب الفرد اليومي من البروتين حسب نوع الأسرة
2362	Average daily per capita protein from animal production	متوسط نصيب الفرد اليومي من البروتين من المنتجات الحيوانية
2325	Average daily per capita protein from plant production	متوسط نصيب الفرد اليومي من البروتين من المنتجات النباتية
2321	Average daily per capita fats by household type	متوسط نصيب الفرد اليومي من الدهون حسب نوع الأسرة
2328	Average daily per capita fats from animal production	متوسط نصيب الفرد اليومي من الدهون من المنتجات الحيوانية
2361	Average daily per capita fats from plant production	متوسط نصيب الفرد اليومي من الدهون من المنتجات النباتية
2327	Average daily per capita calories from animal production	متوسط نصيب الفرد اليومي من السعرات الحرارية من المنتجات الحيوانية
2363	Average daily per capita calories from plant production	متوسط نصيب الفرد اليومي من السعرات الحرارية من المنتجات النباتية

S.V. Code	Indicator Name	اسم المؤشر
2326	Average daily per capita calories from plant production by household type	متوسط نصيب الفرد اليومي من السعرات الحرارية من المنتجات النباتية حسب نوع الأسرة
2365	Daily per capita energy, proteins and fats	نصيب الفرد اليومي من الطاقة، البروتينات والدهون
2366	Daily per capita energy, proteins and fats by household type	نصيب الفرد اليومي من الطاقة، البروتينات والدهون حسب نوع الأسرة
1920	Food consumption patterns	نمط الاستهلاك الغذائي

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Comments (if any):

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